

A Trust-Evaluation-Enhanced Blockchain-Secured Industrial IoT System

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Abstract—The Industrial Internet of Things (IIoT), allowing direct wireless communications between industrial machines and people, is the key component of Industry 4.0. By grouping several IIoT devices (IIoT-Ds) into IIoT groups (IIoT-Gs), the IIoT-Ds within one group can share information and deter malicious users from accessing the network by modifying the access control list (ACL) in IIoT-Ds. In an IIoT-G, several IIoT-Ds jointly manage the access control of IIoT sensors. The security of ACL is protected by blockchains deployed in IIoT-Ds. Even secured by blockchains, a blockchain-based IIoT-G (B-IIoT-G) still cannot secure correct access control in a system with more than half of the blockchain-based IIoT-Ds (B-IIoT-Ds) from spreading wrong authorization information. In this article, we propose a B-IIoT-G with trust evaluation for each B-IIoT-D to evaluate the weight in voting for final decisions of authorization. By designing a new voting mechanism with trust evaluation for the access control in B-IIoT-G, the system has a higher probability of making correct authorization even with malicious B-IIoT-Ds. Simulation results have demonstrated that the mechanism with trust evaluation can still make, with a high probability, correct authorization even with more than half of the B-IIoT-Ds in B-IIoT-G being malicious.

Index Terms—Access control, blockchain, Industrial Internet of Things (IIoT), trust evaluation, voting mechanism.

I. INTRODUCTION

IN THE era of Internet of Things (IoT), the number of devices connected to the Internet is increasing year by year. As the primary pillar of Industry 4.0, Industrial IoT (IIoT) is widely used in supermatic factories to control the assembly processes. Most of these IIoT devices (IIoT-Ds) consist of several sensors. However, to make accurate control of different parts of IIoT-Ds, the controller of each IIoT-D needs to collect the data from sensors frequently.¹ By collecting data from different IIoT-Ds, the IIoT server can analyze the data from a global view and give specific feedbacks to different IIoT-Ds, respectively. As a cloud computing server, the IIoT server has been advocated to perform computationally intensive tasks for IIoT-Ds [1]. Thus, IIoT-Ds can obtain the results with little

of their own computing resources by offloading their tasks to IIoT servers.

The cloud servers, empowered with ample computing resources, can complete the computationally intensive tasks much faster than IIoT-Ds do. However, the remotely located cloud servers incur long communication delays between IIoT-Ds and cloud servers [2]. Fog computing has been proposed to reduce the communication delay in offloading computing tasks of IIoT-Ds [3], [4]. Considering the features of an IIoT system, IIoT-Ds should be equipped with adequate computing resource. Thus, IIoT-Ds can be tailored as fog nodes to provide computing resources to analyze data from sensors. By tailoring the fog computing architecture for IIoT-Ds within one service, IIoT-Ds can communicate with each other directly [5]–[7]. Thus, instead of transmitting the collected data to an IIoT server, IIoT-Ds can analyze the collected data by themselves and exchange the results with other IIoT-Ds.

In the IIoT system, IIoT sensors connect with IIoT-Ds via both wireless and wired connections. For sensors with wirelessly connected with IIoT-Ds, IIoT-Ds have to authenticate to the sensors for data transmission. Currently, an authentication, authorization, and accounting (AAA) server is used to manage the authorization for mobile devices, such as IIoT sensors [8]. The time consumed for traditional access control (using tokens assigned by the AAA server to access the network) is increasing rapidly as a growing number of mobile devices flood the network. By delegating the function of authorization from the AAA server to IIoT-Ds, the delay of authorization can be greatly reduced. Since the access requests are distributed to several IIoT-Ds (instead of to one AAA server), the queuing delay at different IIoT-Ds varies widely. It is thus important to design an efficient access control scheme for such a distributed authorization system. The attribute-based access control (ABAC) is a batch process access control scheme used for fog computing [9]. In ABAC, users with correct attributes are authorized to access the system. The efficiency of ABAC is much higher than the traditional scheme, though the accuracy is lower. To balance the security and efficiency, the administrator decided access control scheme [e.g., mandatory access control (MAC)] has been proposed. In a small-scaled network, MAC is more secured than ABAC and is more efficient than the traditional scheme [10]. To increase the security of authorizations, IIoT-Ds are grouped to manage the access control jointly, as shown in Fig. 1. After some of the IIoT-Ds form an IIoT group (IIoT-G), IIoT-Ds within the same group can communicate with each other [11], e.g., sharing authorization

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¹An IIoT-D is composed of two parts: 1) IIoT sensors and 2) IIoT node. The IIoT sensors are the devices that used to collect data. The IIoT node is the controller used to analyze the data from IIoT sensors and exchange data with IIoT servers.

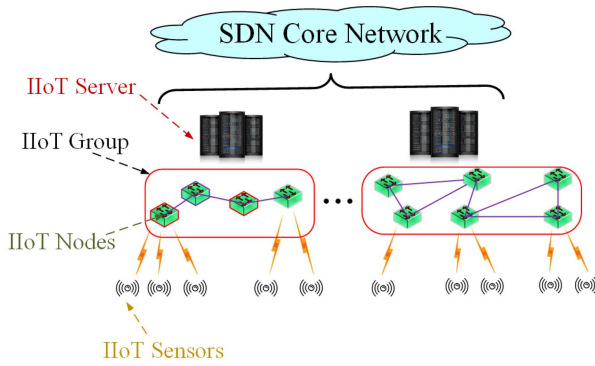


Fig. 1. IIoT system structure.

data and data from sensors.² Thus, one IIoT-G constitutes a peer-to-peer (P2P) network in which MAC is better suited than ABAC.

As a P2P network surrounded with IIoT-Ds, IIoT-Gs are likely exposed to malicious attacks. Thus, a high performance cryptographic algorithm is necessary to secure IIoT-Gs. Asymmetric cryptography (AC) and digital signature (DS) are two popular encryption methods used for user identification in MAC. In small-scale P2P networks, using AC or DS to secure IIoT-Gs does present different drawbacks. By using DS to authenticate identification, a trusted third party (TTP) is needed to generate DS [12]. Yet, a TTP may be far from IIoT-Gs. When a malicious IIoT-D is used to forward DSs from a TTP, the information from the TTP can be changed/monitored/ruined before it reaches the destination [i.e., the man in the middle (MIM) attack] [13]. The MIM attack renders the information from the TTP not trustworthy. Thus, DS is not as secured as it used to be. Different from DS, the AC algorithm does not need a third party to provide keys [14]. The AC algorithm is used within a receiver and several senders (one or more). The receiver generates a pair of keys (i.e., a private key and a public key) at the same time, and broadcasts the public key to the network. The data encrypted by the public key can only be decrypted by the paired private key. The senders who want to send data to the receiver can use the public key to encrypt the data. Since public keys are broadcast to the network by receivers, both legitimate users and malicious users can easily retrieve public keys. If malicious users (e.g., malicious IIoT-Ds) have sufficient computing resources, the private key can be acquired by brute-force cracking the public key in a short time.³ Thus, the information transmitted to the receiver can be decrypted by the malicious users.

In order to increase the robustness of the fog computing system,⁴ we propose a highly robust Blockchain-based IIoT-G (B-IIoT-G) with trust evaluation. The architecture of

B-IIoT-G is illustrated in Fig. 1. Each B-IIoT-G consists of several blockchain-based IIoT-Ds (B-IIoT-Ds). Each B-IIoT-Ds stores the blockchain and generates a dynamic table for trust evaluations of all other B-IIoT-Ds within the same B-IIoT-G. By considering trust values (results of trust evaluation on B-IIoT-Ds) in voting for authorization results, the B-IIoT-G can make correct authorizations even with more than half of the B-IIoT-Ds being malicious.

The remainder of this article is organized as follows. In Section II, we briefly reviewed the related works. In Section III, we introduce the system of B-IIoT-G. In Section IV, we present the system model of B-IIoT-G and formulate the voting mechanism in B-IIoT-G. To improve the probability of making correct authorizations, we propose a voting mechanism based on trust evaluations for B-IIoT-G in Section V. Simulation results to demonstrate the performance of the proposed algorithm is presented in Section VI. A brief conclusion is drawn in Section VII.

II. RELATED WORKS

The complexity of cryptographic algorithms is a key performance metric. To reduce the complexity of cryptographic algorithms, Diro *et al.* [15] proposed a lightweight cryptographic algorithm [based on the elliptic curve cryptography (ECC) algorithm] to handle small sized data. As compared with the RSA algorithm [16], the time required for decryption and encryption of ECC is only 1% of RSA. However, Diro's cryptographic algorithm only works for the information with a particular size (i.e., the data size of all users is the same). The algorithm needs to be redesigned when the data size changes. Thus, it does not perform well for systems with varying sizes of data. Kiviharju [17] discussed the relationship between the security level of fog computing and the length of cryptographic keys, and proposed a variable key length scheme for different scenarios.

Apart from using highly secured cryptographic algorithms, detecting malicious users can secure the network as well [18]–[20]. To detect malicious users, the system needs to monitor the behaviors of all users. However, in a large scale network (e.g., SDN core network), the SDN controller is used to monitor the user behaviors. Since one SDN core network controller needs to monitor all connected users to detect malicious users, the detection task requires tremendous resources [21]. In small-scaled networks (e.g., IIoT-Gs), each controller (IIoT server) only needs to monitor the IIoT-Ds within its coverage. Thus, the required resource for each IIoT server to monitor user behaviors is much less than that for the SDN core network controller. Machine learning can be readily used to detect malicious user behaviors via offline/online training [22]. However, the tremendous required computing resources and long training time have hampered the adoption of machine learning for monitoring malicious users in small-scaled networks. Other than machine learning, several methods [23]–[25] have been proposed to speed up the detection of malicious users. A short detection time can reduce the influence of malicious attacks. Since the malicious users are usually detected after the malicious behaviors have been

²IIoT-Ds in one IIoT-G should be within the coverage of one IIoT server.

³The brute-force cracking is a method to get the private key by trying all possible values of private keys, until the attacker gets the correct one that matches the public key.

⁴In this article, the robustness of a system is reflected by its ability to maintain normal operation even with a number of malicious B-IIoT-Ds. The more malicious B-IIoT-Ds that the system can tolerate without succumbing into malfunction, the more robust this system is.

instigated, the system cannot totally prevent the influence of malicious users [26]. Thus, before detecting malicious users, a system needs to keep working normally under the influence of malicious users. Otherwise, malicious users may disrupt the anomaly detection.

To enhance the robustness of small-scaled networks (i.e., keep the system operating correctly even with malicious users), Sharma *et al.* [27] proposed to employ blockchain in fog/cloud computing to secure the network. As a technology used in the electronic cash system, the blockchain technology can keep the system working normally unless more than 50% of the total computing resources have been compromised [28]. Guo *et al.* [29] proposed a distributed authentication system for the edge computing. In a blockchain secured system, Zhaofeng *et al.* [30] proposed a novel management scheme to make authorizations. Thus, in a blockchain-secured distributed system (e.g., IIoT-G), the blockchains are used to protect the access control list (ACL). The B-IIoT-Ds in one B-IIoT-G share the same blockchain and ACL. Thus, in one B-IIoT-G, the authorizations of IIoT-Ds are voted by all B-IIoT-Ds. By following the majority rule (a fundamental voting mechanism), a B-IIoT-G can keep the correct access control as long as less than 50% of B-IIoT-Ds are malicious. To further increase the system robustness (i.e., keep the system working normally with more malicious B-IIoT-Ds), the trust evaluation is applied in the authorization voting mechanisms. The malicious B-IIoT-Ds are given small voting weights through trust evaluations, and thus the B-IIoT-G can keep working normally even with more than 50% malicious B-IIoT-Ds.

III. BLOCKCHAIN-BASED B-IIoT-G

In a B-IIoT-G, blockchains in B-IIoT-Ds are used to secure the access control. The voting mechanism of blockchains is an efficient way to increase the robustness of B-IIoT-G (i.e., the ability of making correct authorizations with malicious B-IIoT-Ds). To use the blockchain technology, each B-IIoT-D needs to store the ever growing blockchain, which requires a large storage space. In addition, B-IIoT-Ds need to invest huge computing resources to acquire a unique hash value for each block.

To reduce the cost of using blockchains, we design a lightweight blockchain to condense the storage space requirements and reduce the required computing resource simultaneously. Based on the proposed lightweight blockchains, a new voting mechanism with trust evaluation is proposed to enhance the robustness of B-IIoT-G.

A. Blockchain in B-IIoT-G

The internal structure of a block in B-IIoT-G is illustrated in Fig. 2. Each block records all results of IIoT sensors accessing requests (rejected or approved to access the network) and all changes of the ACL in the previous static timeslot [31]. Each requests (including its returned result) or any updates in ACL are viewed as one transaction in the blockchain. The hash value of each transaction forms a Merkle tree (a binary tree with hash function) and the Merkle Root (the root of Merkle tree) is stored in the block header [32]. The ACL Hash is the

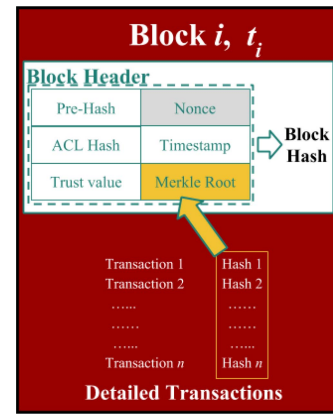


Fig. 2. Structure of a block.

hash value of the ACL at the end of each timeslot, and it can be used to synchronize ACLs in the B-IIoT-G at the beginning of the next timeslot. To combine blocks together to form a chain, a new block includes the hash value of the previous block (i.e., Prehash) in the blockchain. The hash value of each block, [referred to as block hash (BH)], is calculated by using all values in its block header. The “Nonce” is used to ensure that the BH satisfies the following requirements: 1) the BH falls within a subset of all possible results and 2) the BH of a new block should be different from BHs of previous blocks. The first B-IIoT-D that acquires a BH that satisfies the system requirements can broadcast the result to the whole B-IIoT-G, and all other B-IIoT-Ds must follow that BH [33].

As a time-sensitive network, B-IIoT-G does not need to store the blocks generated a long time ago (e.g., a week or more, depending on the system). Thus, the length of a blockchain has to be limited in terms of a static number of blocks⁵ (i.e., the old blocks are automatically deleted when the length reaches the upper bound). A blockchain with a short length cannot trace the ledger to the very beginning, and thus reduces the security level [34]. To deploy a blockchain on a B-IIoT-G, the security level is inevitably compromised owing to the limited storage.

B. Voting Mechanism of B-IIoT-G

The authorizations of IIoT sensors in a B-IIoT-G are based on the voting mechanism used by blockchains. The current voting mechanism for making authorizations used by blockchain follows the majority rule. Thus, a malicious user controls the voting result if over 50% B-IIoT-Ds are hacked by the malicious user. Considering the scale of a B-IIoT-G (with a limit number of B-IIoT-Ds), it is not very hard to hack 51% of B-IIoT-Ds.

To improve accuracy of the voting mechanism, each B-IIoT-D evaluates the trust of all other B-IIoT-Ds within one B-IIoT-G. Based on the trust evaluation, different B-IIoT-Ds assume different weights in voting for authorization. The results of trust evaluation are represented by “trust values.” Each B-IIoT-D stores the trust values of the B-IIoT-Ds within the same B-IIoT-G in its own database. B-IIoT-Ds with a low trust value are given

⁵The “static number” is predefined by the system.

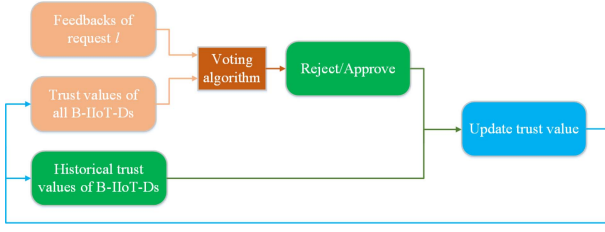


Fig. 3. Decision mechanism.

TABLE I
PARAMETER TABLE

$\mathcal{A}^l$	The matrix of feedbacks to all requests from all B-IIoT-Ds in timeslot l
$\mathcal{G}_i^l$	The matrix of historical trust values of all B-IIoT-Ds recorded by B-IIoT-D i that is used in timeslot l
$\mathcal{V}^l$	The matrix of trust values of all B-IIoT-Ds acquired by different B-IIoT-Ds that is used to make authorizations in timeslot l
$\vec{H}_i^l$	The enhancement vector used by B-IIoT-D i in timeslot l
$\beta^{l,p}$	The authorization result of the p -th request at timeslot l according to correct ACL
$a_{p,i}^l$	The feedback to the p -th request from B-IIoT-D i in timeslot l (approve: $a_{p,i}^l = 1$; reject: $a_{p,i}^l = -1$)
$b_i^{l,p}$	The authorization result acquired by B-IIoT-D i to the p -th request in timeslot l by the EV mechanism
$f_i^{l,p}$	The authorization result acquired by B-IIoT-D i to the p -th request in timeslot l by the TED algorithm
$v_{i,j}^l$	The trust value of B-IIoT-D j acquired by B-IIoT-D i used in timeslot l
$w_{i,j}^l$	The number of wrong authorizations of B-IIoT-D j collected by B-IIoT-D i during timeslot l
r^l	The total number of requests in the l -th timeslot
n	Number of malicious B-IIoT-Ds
k	Number of blocks in the blockchain
y	The correct authorizations ratio threshold
d_1^{th}, d_2^{th}	The thresholds of d , ($0 < d_1^{th} < d_2^{th} < 2$)

a small weight in voting for authorization results. The voting mechanism based on trust values is illustrated in Fig. 3. Each B-IIoT-D collects the feedbacks from all other B-IIoT-Ds in response to a request. The authorization result (the decision based on feedbacks and trust values) is made by combining the trust values and the feedbacks. After each B-IIoT-D gets the authorization result, it updates the trust values of other B-IIoT-Ds by jointly considering the historical trust values, the feedbacks from other B-IIoT-Ds, and the authorization result.

IV. PROBLEM FORMULATION

Fig. 1 shows the system model of a B-IIoT-G. In a B-IIoT-G, the B-IIoT-Ds can communicate with each other directly. We assume that each B-IIoT-D can communicate with any other B-IIoT-D only through traditional switches. Thus, the relationship between each B-IIoT-D is denoted by a direct connection. Each B-IIoT-D can analyze the data collected from IIoT sensors and authenticate IIoT sensors simultaneously. The accessing requests from IIoT sensors are forwarded by the connected B-IIoT-D to the B-IIoT-G.

Let the number of B-IIoT-Ds in one B-IIoT-G be m . Other parameters are defined in Table I. Based on the mechanism of generating new blocks in blockchains, a new block is generated and added to the blockchain at the end of each timeslot.

We assume that there are n malicious B-IIoT-Ds in the system. A malicious B-IIoT-D is defined as a B-IIoT-D with a high probability of giving wrong feedbacks.

$\mathcal{A}^l$ (an $r^l \times m$ matrix) is the matrix of feedbacks for all requests in timeslot l . r^l is the total number of requests in timeslot l . Each row in $\mathcal{A}^l$ stands for a request in timeslot l . Each column stands for a B-IIoT-D that provides feedbacks. Each element in $\mathcal{A}^l$ ($a_{p,i}^l$) stands for the feedback from B-IIoT-D i to request p in timeslot l ($a_{p,i}^l = -1, +1$). Vector $\vec{A}_p^l$ corresponds to the p th row of matrix $\mathcal{A}^l$ and stands for the feedbacks from all B-IIoT-Ds to request p . $\mathcal{V}^l$ (an $m \times m$ matrix) is the matrix of trust values to all B-IIoT-Ds stored in different B-IIoT-Ds during timeslot l , in which each row represents the trust values of all other B-IIoT-Ds perceived and calculated by a particular B-IIoT-D, and each column represents the trust values of a particular B-IIoT-D perceived and calculated by other B-IIoT-Ds. Thus, each element in $\mathcal{V}^l$ ($v_{i,j}^l$) stands for the trust value of B-IIoT-D j stored at B-IIoT-D i ($v_{i,j}^l \in [0, 1]$).

We formulate the problem of maximizing the ratio of malicious B-IIoT-Ds in a normal working B-IIoT-G as follows:

$$\begin{aligned} \mathbf{P0}: \quad & \max_{\mathcal{V}^l} \frac{n}{m} \\ \text{s.t.: } \quad & \text{C1: } \frac{\sum_{p=1}^{r^l} \left| \beta^{l,p} - \vec{A}_p^l \cdot \vec{V}_i^l / \left| \vec{A}_p^l \cdot \vec{V}_i^l \right| \right|}{2 \cdot r^l} \leq y \end{aligned}$$

where n is the number of malicious B-IIoT-Ds and m is the number of all B-IIoT-Ds in one B-IIoT-G. The authorization result to request p from a correct ACL in timeslot l is denoted by $\beta^{l,p}$ ($\beta^{l,p} = 1$ if the authorization result is approval and -1 otherwise). Constraint C1 ensures that the ratio of wrong authorizations (i.e., the result acquired by the voting mechanism is different from the result of a correct ACL) in one timeslot should be smaller than the threshold y . Here, a B-IIoT-G is considered normal if the probability of making wrong decisions is smaller than y .

Thus, we define the vector $\vec{V}_i^l$ as the i th row of matrix $\mathcal{V}^l$, corresponding to the trust values of all B-IIoT-Ds perceived by and stored in B-IIoT-D i in timeslot l . The authorization result of the p th request in timeslot l acquired by B-IIoT-D i is obtained by the inner product of vectors $\vec{V}_i^l$ and $\vec{A}_p^l$ (i.e., when $\vec{V}_i^l \cdot \vec{A}_p^l$ is positive, the authorization result is approval; otherwise, it is rejection).⁶ We further normalize the inner product as $\vec{V}_i^l \cdot \vec{A}_p^l / |\vec{V}_i^l \cdot \vec{A}_p^l|$, so that each normalized authorization result is either $+1$ or -1 . The number of wrong authorizations is then

$$\frac{\sum_{p=1}^{r^l} \left| \beta^{l,p} - \vec{A}_p^l \cdot \vec{V}_i^l / \left| \vec{A}_p^l \cdot \vec{V}_i^l \right| \right|}{2}. \quad (1)$$

The ratio of wrong authorizations in timeslot l is obtained by dividing (1) by the total number of requests r^l , thus resulting in the constraint C1.

V. TRUST-BASED VOTING MECHANISM

The voting mechanism for authorization in a B-IIoT-G is illustrated in Fig. 3. To improve the robustness of a B-IIoT-G, we design a voting mechanism for authorization based on trust

⁶Here, we assume $\vec{V}_i^l \cdot \vec{A}_p^l$ cannot equal to 0.

Algorithm 1 Authorization Mechanism

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1: Input:  $\mathcal{A}^l, \mathcal{V}^l, \mathcal{G}_i^l, d_1^{th}, d_2^{th}, r^l$ 
2: Output:  $\mathcal{V}^{l+1}$ 
3: for all requests in timeslot  $l$  do
4:   Input  $\mathcal{A}^l, \tilde{\mathcal{V}}_i^l, d_1^{th}, d_2^{th}$  and  $w_{i,j}^l$  to Alg. 2 to update  $w_{i,j}^l$ 
     for all B-IIoT-Ds.
5: end for
6:  $\tilde{\mathcal{V}}_i^{l+1} = \{v_{i,1}^{l+1}, v_{i,2}^{l+1}, \dots, v_{i,j}^{l+1}, \dots, v_{i,m}^{l+1}\},$ 
    $v_{i,j}^{l+1} = 1 - (w_{i,j}^l/r^l), \forall j \leq m$ 
7: For all B-IIoT-Ds, input the matrix of historical trust
   values  $\mathcal{G}_i^l$  to Alg. 3
8: Get the enhancement vector  $\tilde{H}_i^l$  and generate the new trust
   value  $\tilde{V}_i^{l+1} = \alpha \cdot \tilde{\mathcal{V}}_i^{l+1} + (1 - \alpha) \cdot \tilde{H}_i^l$  for the next timeslot
9: The trust values for next timeslot ( $\mathcal{V}^{l+1}$ ) are generated by
    $\tilde{V}_i^{l+1}$  from all B-IIoT-Ds.
10: output  $\mathcal{V}^{l+1}$ 

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evaluations (illustrated in Algorithm 1). Algorithm 1 consists of two tasks: 1) a trust evaluation and decision (TED) algorithm (Algorithm 2) and 2) a trust value enhancement (TVE) algorithm (Algorithm 3). Algorithm 2 is invoked for every request (making authorizations and recording the number of wrong authorizations for all B-IIoT-Ds). At the end of each timeslot, Algorithm 2 collects the number of wrong feedbacks ($w_{i,j}^l$) and generates temporary trust values vector for the next timeslot ($\tilde{\mathcal{V}}_i^{l+1}$) by each B-IIoT-D, respectively. Algorithm 3 generates a TVE vector ($\tilde{H}_i^l$) to each B-IIoT-D by giving different weights to the historical trust values ($\mathcal{G}_i^l$) at the end of each timeslot, respectively. By giving different weights (α and $1 - \alpha$) to $\tilde{\mathcal{V}}_i^{l+1}$ and $\tilde{H}_i^l$, the vector of permanent trust values to all B-IIoT-Ds acquired by B-IIoT-D i for the next timeslot ($\tilde{V}_i^{l+1}$) is updated in step 8 in Algorithm 1. The matrix of all trust values for timeslot $l + 1$ ($\mathcal{V}^{l+1}$) is generated by step 9.

A. Trust Evaluation and Decision Algorithm

We propose a TED algorithm to acquire the authorization results to all requests in timeslot l and record the number of wrong authorizations of all B-IIoT-Ds; the details are summarized in Algorithm 2. $a_{p,i}^l$ is the feedback from B-IIoT-D i to request p in timeslot l . $a_{p,i}^l = 1$, when the feedback is approval, otherwise, $a_{p,i}^l = -1$. $v_{i,j}^l$ is the trust value assigned to B-IIoT-D j (evaluated) by B-IIoT-D i in timeslot l ($v_{i,j}^l \in [0, 1]$). The closer $v_{i,j}^l$ is to 1, the more trustworthy B-IIoT-D j is to B-IIoT-D i . To indicate the trust status of B-IIoT-Ds in a B-IIoT-G (e.g., the number of malicious B-IIoT-Ds and trust values of malicious B-IIoT-Ds), we define a parameter to request p in timeslot l ($d_i^{l,p}$) as

$$d_i^{l,p} = \frac{\sum_{j=1, j \neq i}^m |a_{p,i}^l - a_{p,j}^l \cdot v_{i,j}^l|}{m - 1} \quad (2)$$

which represents the trust value status of B-IIoT-G. Note that $d_i^{l,p} \in [0, 2]$. When most of B-IIoT-Ds with high trust values provide the same feedbacks $a_{p,i}^l$, $d_i^{l,p}$ is close to 0. When most of B-IIoT-Ds with high trust values provide feedbacks contrary

Algorithm 2 TED Algorithm

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1: Input:  $\mathcal{A}^l, \tilde{\mathcal{V}}_i^l, d_1^{th}, d_2^{th}, w_{i,j}^l$ 
2: Output:  $w_{i,j}^l$ 
3: for all  $v_{i,j}^l \in \tilde{\mathcal{V}}_i^l$  do
4:   Get the  $d_i^{l,p}$  from Eq. (2) with  $v_{i,j}^l \in \tilde{\mathcal{V}}_i^l$ 
5:   if  $d_1^{th} \leq d_i^{l,p} \leq d_2^{th}$  then
6:     Move the smallest  $v_j$  from  $\tilde{\mathcal{V}}_i^l$  to  $\tilde{\mathcal{V}}_i^l$ 
7:     Restart the Loop (Go to Step 4)
8:   else
9:     End the loop (Go to Step 12)
10:  end if
11: end for
12: if  $d_i^{l,p} < d_1^{th}$  then
13:    $f_i^{l,p} = a_{p,i}^l$ 
14: else
15:    $f_i^{l,p} = -a_{p,i}^l$ 
16: end if
17:  $w_{i,j}^l = w_{i,j}^l + 1$  for all  $j$  with  $f_i^{l,p} = -a_{p,i}^l$ 
18: output  $w_{i,j}^l$ 

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to $a_{p,i}^l$, $d_i^{l,p}$ is close to 2. Thus, when $d_i^{l,p}$ is close to 0 or 2, the B-IIoT-G is likely working normally. When all trust values are small, or the summation of $a_{p,j}^l \cdot v_{i,j}^l$ approximates to 0, $d_i^{l,p}$ is close to 1. Thus, with different values of $d_i^{l,p}$, the trust status of a B-IIoT-G is summarized into two cases.

Case 1: $d_i^{l,p}$ being close to 1 implies that there are many malicious B-IIoT-Ds or the trust values of malicious B-IIoT-Ds are high.

Case 2: $d_i^{l,p}$ being close 0 or 2 implies that there are few malicious B-IIoT-Ds or the trust values of malicious B-IIoT-Ds are small in the B-IIoT-G.

Thus, we define two thresholds d_1^{th} and d_2^{th} ($d_1^{th} < d_2^{th}$) to identify the status of B-IIoT-G. When $d_i^{l,p} \in [d_1^{th}, d_2^{th}]$, we identify the status of B-IIoT-G with case 1. Otherwise, the status of B-IIoT-G aligns with case 2. If $d_i^{l,p} \in [d_1^{th}, d_2^{th}]$, we block the B-IIoT-D with the lowest trust value and calculate $d_i^{l,p}$ again, until $d_i^{l,p} \notin [d_1^{th}, d_2^{th}]$ (steps 5–7). When $d_i^{l,p} \notin [d_1^{th}, d_2^{th}]$, the authorization result to request p is made by steps 12–16. $d_i^{l,p} \in [0, d_1^{th}]$ indicates that most of the B-IIoT-Ds have the same feedbacks as that of B-IIoT-D i (assuming the algorithm is run by B-IIoT-D i). Thus, the authorization result to request p acquired by B-IIoT-D i ($f_i^{l,p}$) is equal to $a_{p,i}^l$ (i.e., $f_i^{l,p} = a_{p,i}^l$). Otherwise, (i.e., $d_i^{l,p} \in (d_1^{th}, 2]$), $f_i^{l,p} = -a_{p,i}^l$. For any B-IIoT-D j with $f_i^{l,p} = -a_{p,j}^l$, $w_{i,j}^l$ recorded by B-IIoT-D i is increased by 1 (step 17). Thus, at the end of timeslot l , the value of $w_{i,j}^l$ stands for the number of B-IIoT-D j with $f_i^{l,p} = -a_{p,j}^l$, recorded by B-IIoT-D i .

Each B-IIoT-D evaluates the trust of all other B-IIoT-Ds (based on their feedbacks in the current timeslot) to update the trust values, at the end of each timeslot. Thus, we can classify the status of each request into six cases. The details of trust evaluations with different cases are shown in Fig. 4. Based on the initial value of $d_i^{l,p}$ [acquired by (2)], six cases

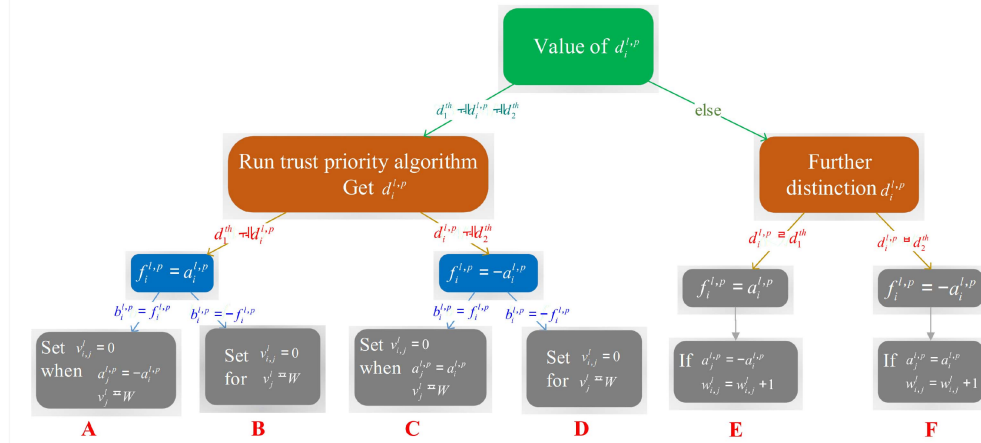


Fig. 4. Trust evaluation and decision (TED).

are grouped into two parts. If $d_i^{l,p} \notin [d_1^{th}, d_2^{th}]$, the number of malicious B-IIoT-Ds is small, or the trust values of malicious B-IIoT-Ds are very low. Thus, the B-IIoT-G does not need to block malicious B-IIoT-Ds (cases E and F). Blocking a B-IIoT-D implies setting the trust value of this B-IIoT-D to 0, and thus feedbacks from this B-IIoT-D cannot influence authorization results. Otherwise, ($d_i^{l,p} \in [d_1^{th}, d_2^{th}]$), there are many malicious B-IIoT-Ds in the B-IIoT-G or some malicious B-IIoT-Ds have high trust values. Thus, some B-IIoT-Ds need to be blocked (cases A, B, C, and D). In cases E and F, B-IIoT-D i records the number of wrong feedbacks from B-IIoT-D j during the current timeslot l in parameter $w_{i,j}^l$. The authorization result to request p in timeslot l acquired by B-IIoT-D i by the TED algorithm is denoted by $f_i^{l,p}$. The voting result following the majority rules ($b_i^{l,p}$) is calculated by (3), which indicates the authorization result to request p acquired by B-IIoT-D i in timeslot l by equally weighted voting (EV) mechanism (the majority rule voting mechanism)

$$b_i^{l,p} = \sum_{j \in V} a_{i,j}^{l,p} / \left| \sum_{j \in V} a_{i,j}^{l,p} \right|. \quad (3)$$

The number of malicious B-IIoT-Ds in cases A–D is many more than those in cases E and F. Cases A and C indicate that feedbacks from malicious B-IIoT-Ds cannot influence the authorization results in the EV mechanism. Thus, we only block those B-IIoT-Ds with the result contradictory to $f_i^{l,p}$, and record the blocked trust values in a temporal set W . Cases B and D indicate that the feedbacks from malicious B-IIoT-Ds can control the authorization results if the EV mechanism or the proposed TED algorithm is used. Thus, in these two cases, we block all B-IIoT-Ds with low trust values. The blocked trust values are recorded in temporal set W .

B. Trust Value Enhancement

Different from Algorithm 2 that is used for each request, the trust values revising algorithm (Algorithm 3) is used at the end of each timeslot.

At the end of each timeslot, Algorithm 2 assigns new trust values to all B-IIoT-Ds for the next timeslot (based on the

Algorithm 3 TVE

- 1: **Input:** $\mathcal{G}_i^l$
- 2: **Output:** $\tilde{H}_i^l$
- 3: Generate the weight vector: $\tilde{N}$
 $\tilde{N} = [1/2^k, 1/2^{k-1}, \dots, 1/2^2, 1/2]$
- 4: $\tilde{H}_i^l = \tilde{N} \cdot \mathcal{G}_i^l$
- 5: **output** $\tilde{H}_i^l$

number of wrong feedbacks of all B-IIoT-Ds). However, generating the trust values only by using the number of wrong feedbacks in the current timeslot may not be accurate enough. Thus, we use Algorithm 3 to generate the new trust values by using historical trust values and the data from Algorithm 2. In Algorithm 3, the number of historical trust values assigned to each B-IIoT-D equals to the number of blocks (k). Thus, the size of the matrix of historical trust values assigned to all B-IIoT-Ds is $k \times m$. We assume that recent historical values are more valuable, and thus we give a higher weight for recent historical values. Thus, we generate a vector $\tilde{N}$ to weigh the historical trust values (a $k \times m$ matrix $\mathcal{G}_i^l$) in step 3. The elements in $\tilde{N}$ are calculated by $\tilde{N}(i) = 1/2^{k-i+1}$ ($\tilde{N}(i)$ is the i th element in $\tilde{N}$), which gives the more recently generated trust values larger weights. The TVE vector ($\tilde{H}_i^l$) is the inner product of the weight vector ($\tilde{N}$) and the historical trust values matrix $\mathcal{G}_i^l$ (step 4).

VI. SIMULATION RESULTS

We conducted all simulations using MATLAB 2019b on Windows 10 operating system to verify the performance of our proposed algorithm. In the simulation, the number of B-IIoT-Ds in the B-IIoT-G is 16 and the total number of timeslots is 40. The malicious B-IIoT-Ds are randomly selected at the beginning of each simulation and do not change for the rest timeslot of the simulation.

The malicious B-IIoT-Ds in the simulation have a high probability (95%) of giving wrong feedbacks to each request. There are 200 requests in each timeslot. Considering the error rate of the network transmission, we let the probability of normal B-IIoT-Ds in sending the wrong feedback be u . The initial trust

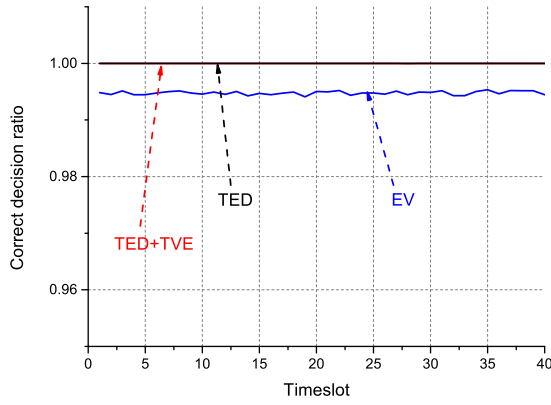


Fig. 5. CDR of TED+TVE, TED, and EV with five malicious B-IIoT-Ds.

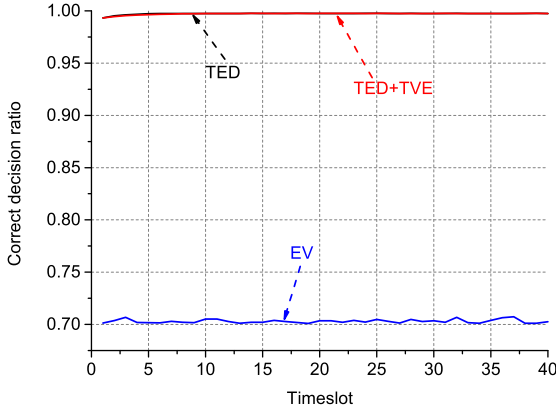


Fig. 6. CDR of TED+TVE, TE and EV with 8 malicious B-IIoT-Ds.

value of all B-IIoT-Ds is set to 1. In the simulation, we compare the correct decision ratio (CDR) of TED+TVE, TED, and EV for different numbers of malicious B-IIoT-Ds. The EV mechanism following the majority rule is used as a reference algorithm. In the TED algorithm, the trust values are only related to $w_{i,j}^t$ in each timeslot. Fig. 5 shows the CDRs of different algorithms with five malicious B-IIoT-Ds. All three algorithms attain stable CDRs (CDR varies little) in different timeslots. The CDRs achieved by TED and TED+TVE are close to 1, much higher than that of EV. In a B-IIoT-G with less than 50% B-IIoT-Ds being malicious, all three algorithms have high CDRs.

The CDRs of three algorithms with eight malicious B-IIoT-Ds (50% B-IIoT-Ds are malicious) are shown in Fig. 6. The CDRs achieved by TED+TVE and TED are still close to 1. However, the CDRs achieved by EV decrease to as low as 70%, implying that the performance of EV is much worse than TED+TVE and TED. Unlike the curves with five malicious B-IIoT-Ds as shown in Fig. 5, the curves with eight malicious B-IIoT-Ds take a few more timeslots to converge. The stable CDRs achieved by TED+TVE and TED are approximately the same. However, TED takes fewer timeslots than TED+TVE to reach the stable CDR.

The CDRs achieved by the three algorithms with ten malicious B-IIoT-Ds (more than half B-IIoT-Ds are malicious) are shown in Fig. 7. The CDR achieved by EV is as low as 0.2, implying that EV cannot make reliable authorizations.

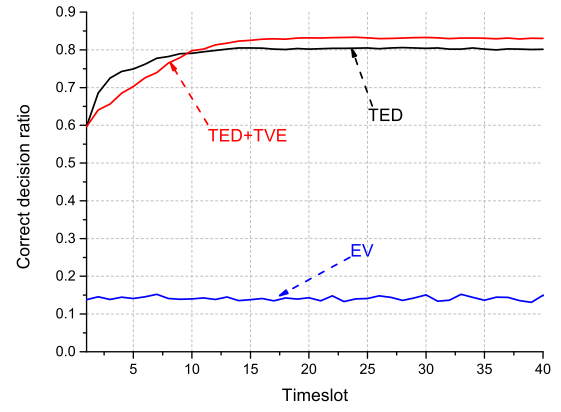


Fig. 7. CDR of TED+TVE, TED and EV with 10 malicious B-IIoT-Ds.

In a system with more than half of B-IIoT-Ds being malicious, TED+TVE and TED achieve much higher CDRs than EV does. Both TED+TVE and TED have low CDRs at the beginning of the simulation. With the system running, TED and TED+TVE can automatically adjust the trust values of malicious B-IIoT-Ds and reduce the influence of the feedbacks from malicious B-IIoT-Ds. Since TED does not use historical trust values of B-IIoT-Ds to generate the trust values for the next timeslot, the CDR of TED increases faster than TED+TVE. However, by using the historical trust values, TED+TVE can make more reasonable trust values for the next timeslot and achieve a higher CDR than TED does (even TED+TVE takes more timeslots).

Based on the simulation results, we have demonstrated that in a B-IIoT-G with few malicious B-IIoT-Ds (less than half B-IIoT-Ds), TED+TVE, TED, and EV achieve similar CDRs. In a B-IIoT-G with many malicious B-IIoT-Ds (especially over 50% of B-IIoT-Ds are malicious), CDRs achieved by both TED+TVE and TED are significantly larger than that acquired by EV. Even the CDR achieved by TED+TVE is higher than that by TED, TED+TVE takes more time to converge than TED does. When nearly 50% of B-IIoT-Ds are malicious, the CDRs of TED+TVE and TED are almost the same (TED also takes fewer timeslots to coverage).

VII. CONCLUSION

In this article, we have proposed a B-IIoT-G architecture by using blockchain to increase the security level. A voting mechanism for authorization based on trust evaluation (consists of TE and TED) has been proposed to increase the robustness of B-IIoT-G. As compared to the voting mechanism following majority rules (EV algorithm), the proposed voting mechanism (TED+TVE) can achieve a higher probability of making correct authorizations. In a B-IIoT-G with more than half B-IIoT-Ds being malicious, TED+TVE can still achieve correct authorizations with a probability of 83%. By analyzing the CDR curves of TED+TVE and TED, we have demonstrated that by generating the new trust values with historical trust values, TED+TVE can achieve a higher CDR than TED, especially with a high ratio of B-IIoT-Ds being malicious. However, TED takes less time to converge and achieves similar CDRs with TED+TVE, when the ratio of malicious B-IIoT-Ds

is near 50%. Thus, we can conclude that TED+TVE is more suitable than TED for a B-IIoT-G with a high ratio of malicious B-IIoT-Ds. In a B-IIoT-G with few B-IIoT-Ds (near or less than 50% of B-IIoT-Ds being malicious), TED converges faster than TED+TVE and both achieve similar CDRs at the same time.

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